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GraphBin-Tk: assembly graph-based metagenomic binning toolkit

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Summary

The study of genetic material directly obtained from natural environments, termed metagenomics, offers valuable insights into microbial communities and their impact on human health and environmental dynamics (Edwards et al., 2013; Pargin et al., 2023). Once the genetic material is extracted, sequenced to obtain reads and assembled to obtain contigs, a process known as metagenomic binning is used to cluster contigs into bins that represent different taxonomic groups which results in draft microbial genomes or metagenome-assembled genomes (MAGs) (V. Mallawaarachchi et al., 2024). Several automated metagenomic binning tools incorporating novel computational methods have been introduced (Alneberg et al., 2014; Chandrasiri et al., 2022; D. D. Kang et al., 2019; Pan et al., 2023; Wu et al., 2015; Xue et al., 2022, 2024) which have led to the discovery and characterisation of many novel micro-organisms (Brooks et al., 2017; L. Kang et al., 2024).

Conventional metagenomic binning tools make use of features such as nucleotide composition and abundance information of contigs, yet find it challenging to bin sequences of closely related species and sequences that have noisy features. Binning tools, such as MetaCoAG (V. Mallawaarachchi & Lin, 2022a, 2022b) that use metagenome assembly graphs (a structure containing the connectivity information of contigs) are gaining popularity due to their improved binning results over conventional binning methods. Moreover, assembly graph-based bin refinement tools such as GraphBin (V. Mallawaarachchi et al., 2020) and GraphBin2 (V. G. Mallawaarachchi et al., 2020, 2021) have been introduced to refine binning results from existing binning tools. Yet, these tools exist as individual software and running them individually can be complex, time-consuming and less accessible. Here we present GraphBin-Tk, an assembly graph-based metagenomic binning tool that combines the capabilities of MetaCoAG, GraphBin and GraphBin2, along with additional pre-processing and post-processing functionality into one comprehensive toolkit. GraphBin-Tk is hosted at https://github.com/metagentools/gbintk.

Statement of need

It is crucial to obtain accurate binning results in metagenomic studies to understand the composition and functional potential of microbial communities. Conventional binning methods mainly rely on two features of contigs; 1) nucleotide composition, represented as normalised frequencies of oligonucleotides (short substrings of a particular length) and 2) abundance, the average number of reads that covers each nucleotide base of the contig (V. Mallawaarachchi et al., 2024). Previous studies have found that these tools face several challenges when binning complex datasets (V. Mallawaarachchi et al., 2024):

- Contigs of closely related species can be merged, producing contaminated bins.
- Contigs having features that deviate from their constituent genomes (e.g., the presence of protein-coding regions and repeat regions can lead to deviated nucleotide and abundance features) can be placed in incorrect bins.
- Contigs that are too short (e.g., shorter than 1000 base pairs) can be discarded during binning as they may not capture enough genomic signatures. Such short sequences can contain important regions such as repeats.
- Contigs shared among different genomes are only placed in the bin of the most representative genome.

GraphBin-Tk enhances binning accuracy and addresses the aforementioned challenges by integrating the capabilities of GraphBin (V. Mallawaarachchi et al., 2020), GraphBin2 (V. G. Mallawaarachchi et al., 2020, 2021) and MetaCoAG (V. Mallawaarachchi & Lin, 2022a, 2022b), which leverage the connectivity information of the assembly graph. GraphBin-Tk unifies these state-of-the-art binning solutions in a comprehensive toolkit for metagenomic binning and refinement as shown in Figure 1. It provides users with a more comprehensive set of features and capabilities, enabling them to perform a wider range of tasks related to metagenomic binning without needing to install and execute additional software. GraphBin-Tk also eliminates any compatibility issues that may arise from running separate binning-related software and enhances the user experience by making the software easier to learn and use.

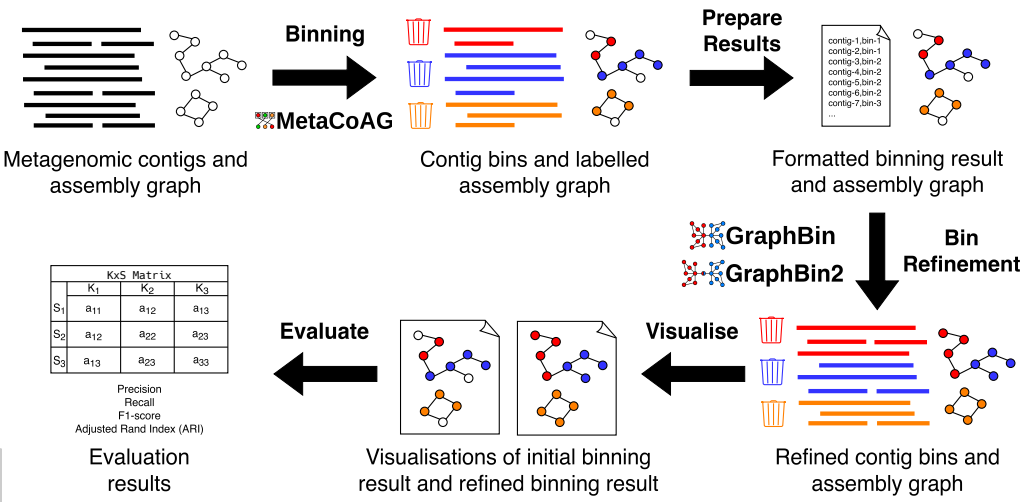

Figure 1: Example binning workflow using tools available from GraphBin-Tk.

GraphBin-Tk can perform stand-alone metagenomic binning using MetaCoAG and bin refinement using either GraphBin or GraphBin2. Additionally, pre- and post-processing functionalities to run these tools and analyse the produced results are included. GraphBin-Tk supports metagenome assemblies generated from three popular metagenome assemblers; metaSPAdes (Nurk et al., 2017) and MEGAHIT (Li et al., 2015) for short-read sequencing data and metaFlye (Kolmogorov et al., 2020) for long-read sequencing data. GraphBin-Tk can be launched using the command `gbintk`. The following subsections explain the subcommands provided in GraphBin-Tk.

`metacoag`

Tool/processing function

A user can start the analysis by running the `metacoag` subcommand to bin a metagenomic dataset using the metagenomic binning tool MetaCoAG (V. Mallawaarachchi & Lin, 2022a, 2022b) and obtain contig bins (or MAGs) as shown in Figure 1.

Inputs

The following inputs are required to run the `metacoag` subcommand.

- Contigs file
- Assembly graph file(s)
- A delimited file containing the contig identifier and its average read coverage for each contig - can be obtained by running a read coverage calculation tool such as CoverM (Woodcroft & Newell, 2017) or Koverage (Roach et al., 2024)

The assembly graph files can vary depending on the assembler used to generate the contigs. The metaSPAdes version requires the assembly graph file in `.gfa` format and the paths file corresponding to the contigs file in `.paths` format. The MEGAHIT version requires the assembly graph file in `.gfa` format. The metaFlye version requires the assembly graph file `assembly_graph.gfa` and the paths file `assembly_info.txt` from the final assembly output.

Outputs

The following outputs will be generated by the `metacoag` subcommand.

- A delimited text file containing the contig identifier and bin identifier for each binned contig
- `.fasta` files of the identified bins

`prepare`

Tool/processing function

If a delimited text file is not available, the initial binning result can be formatted using the `prepare` subcommand into a delimited text file that represents each contig and its bin identifier. This function allows users to format binning results from any metagenomic binning tool.

Inputs

The directory containing the initial binning is required to run the `prepare` subcommand.

Outputs

The `prepare` subcommand will generate a delimited text file such as `.csv` or `.tsv` containing the contig identifier and bin identifier for each contig in the binning result.

`graphbin`

Tool/processing function

A formatted initial binning result from the `prepare` subcommand can be improved by providing it to GraphBin (V. Mallawaarachchi et al., 2020) using the subcommand `graphbin` (Figure 1).

Inputs

The following inputs are required to run the `graphbin` subcommand.

- Contigs file
- Assembly graph file(s) - can vary depending on the assembler used (refer to inputs under `metacoag`)
- A delimited text file containing the initial binning result

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Outputs

The following outputs will be generated by the `graphbin` subcommand.

- A delimited text file containing the contig identifier and bin identifier for each binned contig
- `.fasta` files of the refined bins

`graphbin2`

Tool/processing function

A formatted initial binning result from the `prepare` subcommand can be improved by providing it to GraphBin2 (V. G. Mallawaarachchi et al., 2020, 2021) using the subcommand `graphbin2` (Figure 1).

Inputs

The following inputs are required to run the `graphbin2` subcommand.

- Contigs file
- Assembly graph file(s) - can vary depending on the assembler used (refer to inputs under `metacoag`)
- A delimited text file containing the initial binning result
- A delimited file containing the contig identifier and its average read coverage for each contig - can be obtained by running a read coverage calculation tool (refer to inputs under `metacoag`)

Outputs

The following outputs will be generated by the `graphbin2` subcommand.

- A delimited text file containing the contig identifier and bin identifier for each binned contig
- `.fasta` files of the refined bins

`visualise`

Tool/processing function

The initial binning result and the refined binning result can be visualised on the assembly graph using the `visualise` subcommand (Figure 1). Users can generate images in different formats such as `png`, `eps`, `pdf` and `svg`, and customise the dimensions of the images.

Inputs

The following inputs are required to run the `visualise` subcommand.

- Contigs file
- Assembly graph file(s) - can vary depending on the assembler used to generate the contigs (refer to inputs under `metacoag`)
- A delimited text file containing the initial binning result
- A delimited text file containing the refined binning result

Outputs

The following outputs will be generated by the `visualise` subcommand.

- Figure of the assembly graph with the initial binning result
- Figure of the assembly graph with the refined binning result

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An example visualisation for the Sim-5G+metaSPAdes dataset (V. G. Mallawaarachchi et al., 2020, 2021) containing five bacterial species is shown in Figure 2.

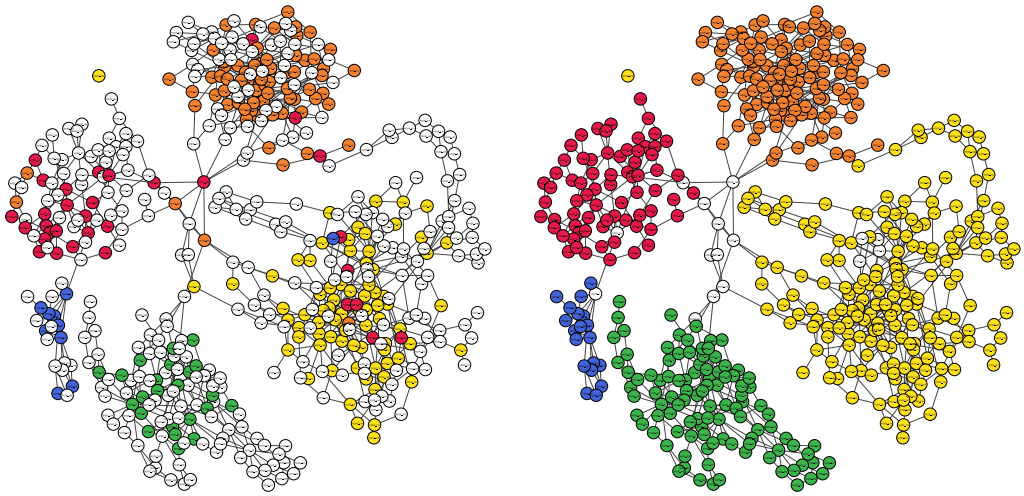

Figure 2: Visualisation of the assembly graph with the initial binning result from MetaCoAG (left) and final binning result from GraphBin (right) for the Sim-5G+metaSPAdes dataset. The vertices represent contigs and edges represent connections in the assembly graph. The five colours represent the five bins and the white vertices represent unbinned contigs.

`evaluate`

Tool/processing function

The produced binning results can be evaluated using the `evaluate` subcommand by providing the ground truth bins of contigs (Figure 1). This evaluation is possible only for simulated or mock metagenomes where the ground truth genomes of contigs are known. GraphBin-Tk uses the four common metrics 1) precision, 2) recall, 3) F1-score and 4) Adjusted Rand Index (ARI) that have been used in previous binning studies (Alneberg et al., 2014; V. Mallawaarachchi et al., 2020; Meyer et al., 2018). These metrics are calculated as follows. The binning result is denoted as a $K \times S$ matrix with K number of bins and S number of ground truth taxa. In this matrix, the element a_{ks} denotes the number of contigs binned to the kth bin and belongs to the sth taxa. U denotes the number of unbinned contigs and N denotes the total number of contigs. Following are the equations used to calculate the evaluation metrics.

Precision $= \frac{\sum_k max_s\{a_{ks}\}}{\sum_k \sum_s a_{ks}}$

Recall $= \frac{\sum_s max_k\{a_{ks}\}}{(\sum_k \sum_s a_{ks}+U)}$

F1-score $= 2 \times \frac{Precision \times Recall}{Precision + Recall}$

ARI $= \frac{\sum_{k,s}\binom{a_{ks}}{2} - t_3}{\frac{1}{2}(t_1+t_2)-t_3}$ $where\ t_1 = \sum_k \binom{\sum_s a_{ks}}{2},\ t_2 = \sum_s \binom{\sum_k a_{ks}}{2},\ and\ t_3 = \frac{t_1 t_2}{\binom{N}{2}}$

Inputs

The following inputs are required to run the `evaluate` subcommand.

- A delimited text file containing the ground truth
- A delimited text file containing the binning result

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Outputs

A text file containing the $K \times S$ matrix and the calculated evaluation metrics will be generated by the `evaluate` subcommand. These metrics can be plotted for comparison between the initial binning result and the refined binning result using custom code as shown in Figure 3.

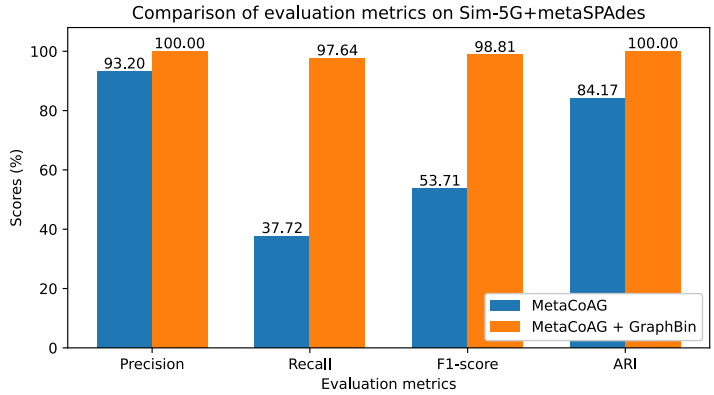

Figure 3: Comparison of evaluation metrics for the intiial binning result from MetaCoAG and the refined binning result from GraphBin for the Sim-5G+metaSPAdes dataset.

Please refer to the original publications of GraphBin (V. Mallawaarachchi et al., 2020), GraphBin2 (V. G. Mallawaarachchi et al., 2020, 2021) and MetaCoAG (V. Mallawaarachchi & Lin, 2022a, 2022b) for detailed benchmarking results of each tool.

Availability

GraphBin-Tk is distributed as a Conda package available in the Bioconda channel (Grüning et al., 2018) at https://anaconda.org/bioconda/gbintk. GraphBin-Tk is also available as a Python package on PyPI at https://pypi.org/project/gbintk. The source code is available on GitHub at https://github.com/metagentools/gbintk and features continuous integration, testing coverage, and continuous deployment using GitHub actions. Detailed documentation and example usage can be found at https://gbintk.readthedocs.io/.

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